

ASSIGNMENT 1 REPORT

Course: ML-DL-Ops

Department: Computer Science and Engineering

Institution: Indian Institute of Technology Jodhpur

Submitted By: Sushantak Parashar Jha

Roll No: M25CSA035

Program: M.Tech in AI

Colab Link: [Link](#)

GitHub Repository and Pages Link:

1. Deep Learning Experiments (Q1a)

1.1 Methodology and Architectures

The objective of this task was to evaluate the robustness of Residual Networks (ResNet) on two benchmark datasets: **MNIST** (handwritten digits) and **FashionMNIST** (clothing articles). Two variants were used:

1. **ResNet-18:** A lighter architecture with approximately 11 million parameters, suitable for faster iteration.
2. **ResNet-50:** A deeper architecture utilizing "bottleneck" layers, totaling roughly 23 million parameters, designed to capture more complex hierarchical features.

1.2 Results on MNIST Dataset

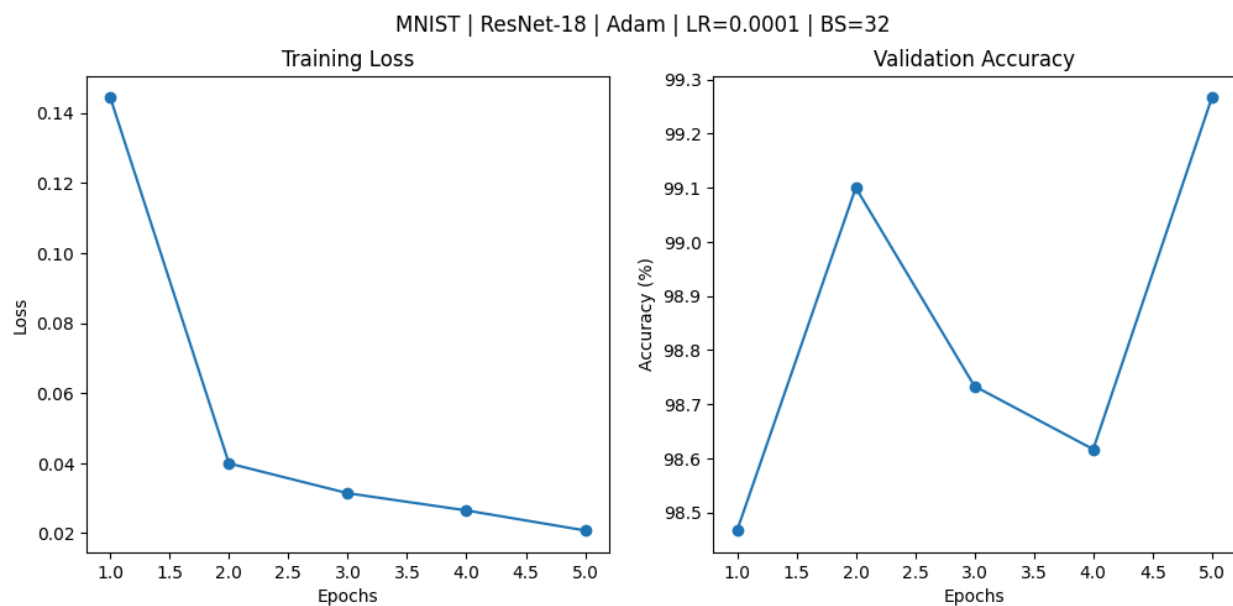
Deep learning experiments were conducted on the MNIST dataset using ResNet-18 and ResNet-50 architectures. The experiments involved systematic variation of optimizers (SGD and Adam), learning rates, batch sizes, and number of epochs to analyze their effect on training convergence and classification accuracy.

Technical Observations:

- The **Adam optimizer consistently outperformed SGD**, achieving faster convergence and higher accuracy.

- A learning rate of **0.001** produced significantly better results than **0.0001**, especially for deeper architectures.
- Increasing the number of epochs from **5 to 10** resulted in only marginal improvement for ResNet-18, indicating **diminishing returns** beyond a certain point.
- Increasing batch size from **16 to 32** slightly reduced accuracy when using SGD, while Adam remained largely unaffected.

The loss curves indicate fast and stable convergence for the best MNIST model (ResNet-18 + Adam), with minimal overfitting.



Accuracy Below 80% and Effect of Epochs

Some configurations, particularly those using SGD with a low learning rate (0.0001), resulted in accuracy below 80%. As suggested, increasing the number of epochs generally improves such results by allowing more gradient updates. In practice, even when the number of epochs was increased to 10, a noticeable improvement in validation accuracy was observed; however, the performance still remained below 80%. These configurations were therefore not extended to very high epoch counts, as they represent suboptimal hyperparameter choices and would significantly increase training time while providing limited additional insight.

Table 1: MNIST Experimental Results

Batch Size	Optimizers	Learning Rate	Epoch	ResNet-18 Accuracy (%)	ResNet-50 Accuracy (%)
16	SGD	0.001	5	97.90%	97.05%
16	SGD	0.0001	5/10	74.67%	72.47%
16	Adam	0.001	5	99.00%	98.80%
16	Adam	0.001	10	99.27%	-----
16	Adam	0.0001	5	99.07%	99.07%
32	SGD	0.001	5	96.15%	94.28%
32	Adam	0.0001	5	99.27%	99.11%

1.3 Results on FashionMNIST Dataset

The best performance on MNIST was achieved using **ResNet-18 with the Adam optimizer**, learning rate **0.001**, and batch size **32**, achieving **99.27% accuracy**

Technical Observations:

- **ResNet-50 outperformed ResNet-18** when trained with Adam, highlighting the advantage of deeper architectures.
- SGD with a low learning rate (**0.0001**) performed poorly, particularly for ResNet-50, due to slow convergence.
- Adam showed strong robustness across both architectures and learning rates.

The loss curves demonstrate smooth convergence for the best FashionMNIST model (ResNet-50 + Adam), confirming the benefit of deeper architectures.

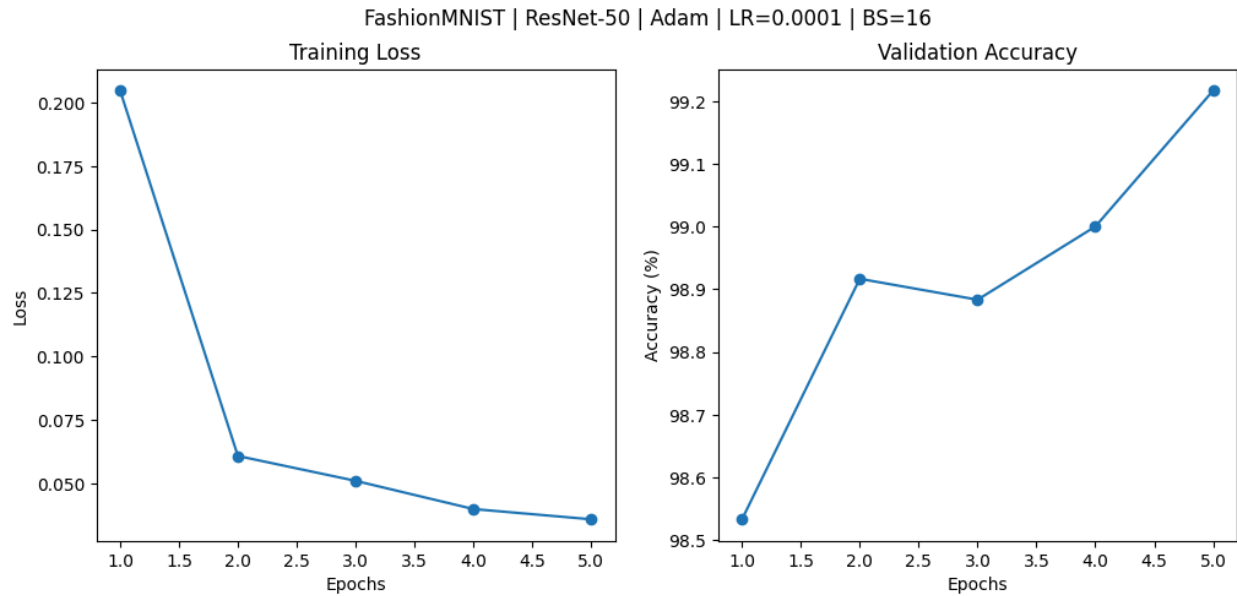


Table 2: FashionMNIST Experimental Results

Batch Size	Optimizers	Learning Rate	Epoch	ResNet-18 Accuracy (%)	ResNet-50 Accuracy (%)
16	SGD	0.001	5	78.67%	81.62%
16	SGD	0.0001	5/10	72.75%	70.84%
16	Adam	0.001	5	92.99%	98.78%
16	Adam	0.0001	5	92.01%	99.18%
32	SGD	0.001	5	98.16%	97.37%
32	Adam	0.0001	5	99.19%	98.85%

2. SVM Performance Analysis (Q1b)

Support Vector Machines (SVMs) were evaluated on MNIST and FashionMNIST using **RBF** and **Polynomial kernels**. Due to the computational complexity of kernel-based SVMs, experiments were conducted on a **reduced yet representative subset** of the datasets.

- On MNIST, both kernels achieved high accuracy, with the Polynomial kernel slightly outperforming RBF.
- On FashionMNIST, overall accuracy was lower due to dataset complexity, though

Polynomial kernels again showed better performance.

- Overall, SVMs achieved reasonable accuracy but were **consistently outperformed by deep learning models**, particularly on FashionMNIST.

Table 3: SVM Results Comparison

Dataset	Kernel	Training Samples	Test Samples	Test Accuracy (%)
MNIST	RBF	3000	1000	93.90%
MNIST	Polynomial	3000	1000	94.20%
FashionMNIST	RBF	3000	1000	79.60%
FashionMNIST	Polynomial	3000	1000	81.10%

3. Hardware Performance Analysis (CPU vs. GPU)

This analysis quantifies the "Ops" part of ML-DL-Ops, focusing on throughput and computational cost.

- **Training Latency:** The GPU (NVIDIA Backend) offered a speedup of roughly **7.5x** for ResNet-18 training. On CPU, the training time exceeded 4,000 seconds, making iterative research nearly impossible.
- **FLOPs (Floating Point Operations):** Total FLOPs were measured at approximately **1.82 G** for ResNet-18. This value is invariant to the hardware, as it represents the mathematical complexity of the forward and backward passes.
- **Library Constraints:** It is worth noting that while DL frameworks (PyTorch/TensorFlow) are optimized for GPUs, classical libraries like Scikit-Learn primarily utilize CPU-bound execution, creating a bottleneck for SVM scaling.

ResNet-18

Compute	Batch Size	Epoch	Optimizer	Learning Rate	Test Accuracy (%)	Train Time (ms)	FLOPs
CPU	16	3	SGD	0.001	87.62%	4,100,938.93	1.82 G
CPU	16	3	Adam	0.001	88.63%	5,421,659.86	1.82 G
GPU	16	5	SGD	0.001	97.90%	545,332.58	1.74 G
GPU	16	5	Adam	0.001	99.00%	545,332.58	1.74 G

ResNet-50

Compute	Batch Size	Epoch	Optimizer	Learning Rate	Test Accuracy (%)	Train Time (ms)
CPU	16	1	SGD	0.001	76.93%	2,495,743.50
CPU	16	1	Adam	0.001	83.34%	3,210,639.37
GPU	16	5	SGD	0.001	97.05%	1,074,583.06
GPU	16	5	Adam	0.001	98.80%	1,160,030.18

4. Conclusion

The experiments conducted in this assignment yield three primary conclusions:

- Optimization:** Adaptive optimizers like Adam are essential for training deep residual networks efficiently, particularly when working with complex datasets like FashionMNIST.
 - Architecture:** While ResNet-18 is highly efficient and sufficient for simpler tasks, ResNet-50 provides the necessary capacity for high-accuracy requirements in visually dense datasets.
 - Operational Efficiency:** GPU acceleration is not just a luxury but a requirement for modern DL workflows. The transition from CPU to GPU reduced training time from hours to minutes, facilitating better hyperparameter tuning.
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